FIGURE · ALGORITHM 5

Secure Aggregation and Momentum-Based Reference Update

A geometric view of the final round update — accepted encrypted coordinates are homomorphically aggregated, only aggregate coordinates are decrypted by the TKA, the averaged global gradient updates the model, and a momentum-normalized trusted reference $R^{(t+1)}$ is stored for the next round.

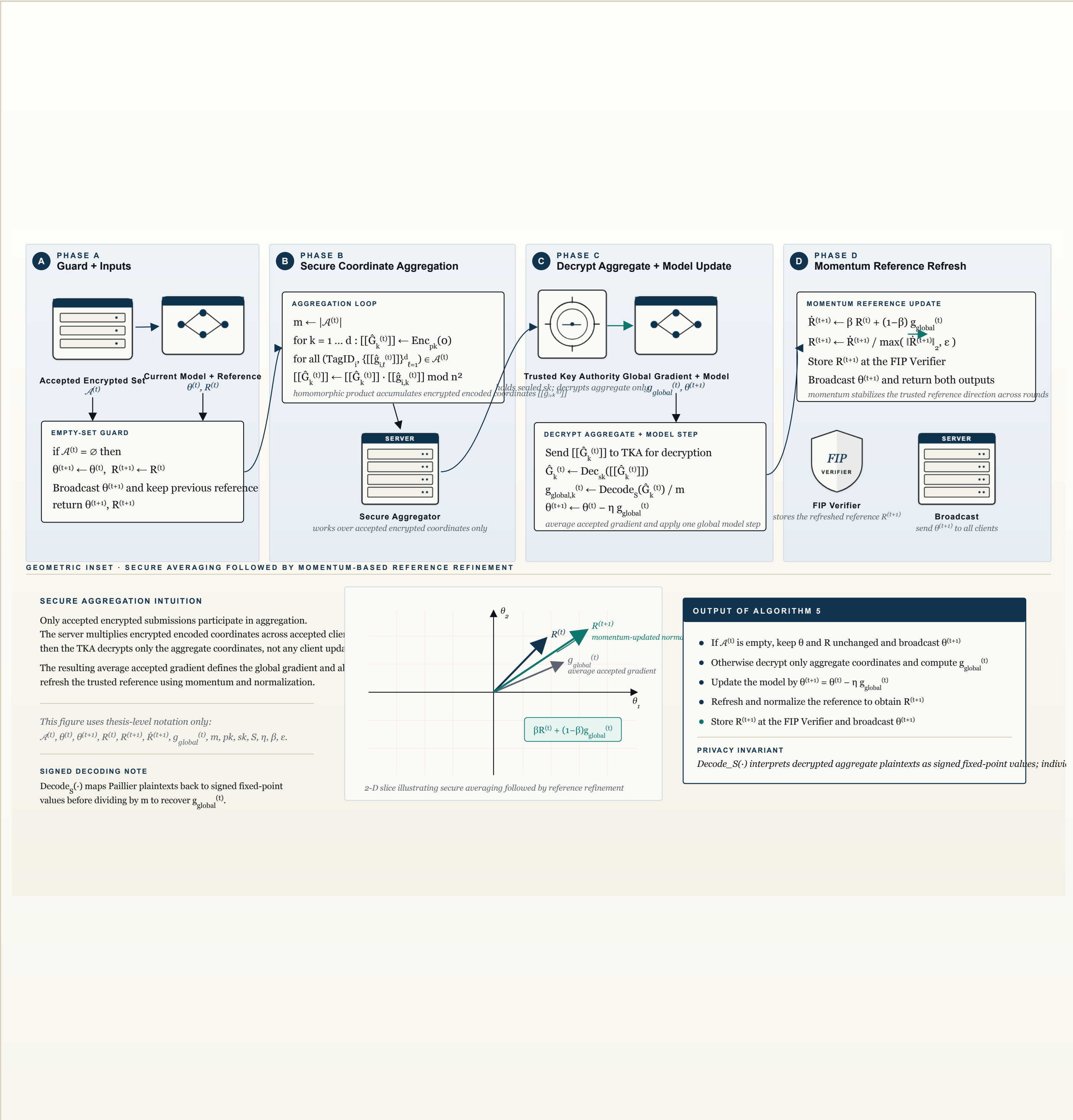

Figure 5. Algorithm 5. If the accepted encrypted set $\mathcal{A}^{(t)}$ is empty, the server keeps $\theta^{(t+1)} \leftarrow \theta^{(t)}$ and $R^{(t+1)} \leftarrow R^{(t)}$, broadcasts the unchanged model, and returns. Otherwise, for each coordinate k, the server homomorphically aggregates the accepted encrypted encoded values into $[[\hat{G}_k^{(t)}]]$ and sends only these aggregate ciphertexts to the TKA. After decryption and signed fixed-point decoding, the average accepted gradient $g_{global}^{(t)}$ is recovered and used to update the model by $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta g_{global}^{(t)}$. The trusted reference is then refreshed as $\dot{R}^{(t+1)} \leftarrow \beta R^{(t)} + (1-\beta) g_{global}^{(t)}$ and normalized to $R^{(t+1)}$ using $\max(\|\dot{R}^{(t+1)}\|_2, \varepsilon)$. The inset illustrates secure averaging followed by momentum-based reference refinement.